

Suppose $X \sim \text{Binomial}(10, \frac{1}{2})$ and $Y \sim \text{Binomial}(11, \frac{1}{2})$, where X and Y are independent random variables. Then $P(X < Y)$ is

- A. less than $\frac{1}{2}$
- B. equal to $\frac{1}{2}$
- C. greater than $\frac{1}{2}$ but less than or equal to $\frac{10}{11}$
- D. greater than $\frac{10}{11}$

A discrete random variable X has PMF as given below where a, b, c are constants.

x	1	2	3	4
$P(X = x)$	a	b	c	0.3

The CDF $F_X(x)$ is given below

x	1	2	3	4
$F_X(x)$	0.2	0.6	0.7	d

Find the value of $a + b + c + d$.

Let X_1, X_2, X_3, X_4, X_5 be i.i.d. random variables having a continuous distribution function. Then

$$P(X_1 > X_2 > X_3 > X_4 > X_5 \mid X_1 = \max(X_1, X_2, X_3, X_4, X_5))$$

is

- A. $1/4$
- B. $1/5$
- C. $1/4!$
- D. $1/5!$

Suppose (X, Y) follows a bivariate normal distribution with means μ_1, μ_2 , standard deviations σ_1, σ_2 , and correlation coefficient ρ , where all the parameters are unknown.

Then testing the null hypothesis

$$H_0 : \sigma_1 = \sigma_2$$

is equivalent to testing the independence of:

- A. X and Y
- B. X and $X - Y$
- C. $X + Y$ and Y
- D. $X + Y$ and $X - Y$

Let $X_1, X_2, \dots$ be a sequence of independent normally distributed random variables with mean 1 and variance 1. Let N be a Poisson random variable with mean 2, independent of $X_1, X_2, \dots$.

Then, the variance of

$$X_1 + X_2 + \dots + X_{N+1}$$

is:

- A. 3
- B. 4
- C. 5
- D. 9

In a random experiment, a fair coin is tossed once. Then, an unbiased six-faced die is rolled N times, where

$$N = \begin{cases} 100, & \text{if Head appears,} \\ 101, & \text{if Tail appears.} \end{cases}$$

Let Y denote the total number of times 6 appears out of N . Then $P(\text{Head} \mid Y = 15)$ equals:

- A. $\frac{516}{1021}$
- B. $\frac{505}{1021}$
- C. $\frac{201}{1021}$
- D. $\frac{1000}{1021}$

Let X be a uniformly distributed random variable with mean $\mu_X = 15$ and variance $\sigma_X^2 = \frac{25}{3}$.

Calculate $P(X > 17)$.

Suppose that X is a random variable such that

$$P(X \in \{0, 1, 2\}) = 1.$$

If for some constant c ,

$$P(X = i) = c P(X = i - 1), \quad i = 1, 2,$$

then the value of $E[X]$ is

A. $\frac{1}{1+c+c^2}$

B. $\frac{c+2c^2}{1+c+c^2}$

C. $\frac{c+c^2}{1+2c}$

D. $\frac{3c}{1+2c}$

Let X and Y be two random variables with joint PMF $f_{X,Y}(x,y)$ given below. Calculate covariance between X and Y .

$Y \backslash X$	1	2	3
1	0.25	0.25	0
2	0	0.25	0.25

Consider events $B_1, B_2, \dots, B_k$ that form a partition of the sample space S . For $i = 1, \dots, k$, let $\Pr(B_i)$ denote the prior probability of B_i . There is another event A such that $\Pr(A) > 0$. Let $\Pr(B_i | A)$ denote the posterior probability of B_i given that the event A has occurred.

Suppose that

$$\Pr(B_1 | A) < \Pr(B_1).$$

Which of the following statements are correct? (Multiple options may be correct.)

- A. $\sum_{i=1}^k \Pr(B_i) = 1$
- B. $\sum_{i=1}^k \Pr(B_i | A) = 1$
- C. There exists at least one value of i , $i = 2, \dots, k$, such that $\Pr(B_i | A) > \Pr(B_i)$
- D. For all $i = 2, \dots, k$, $\Pr(B_i | A) \leq \Pr(B_i)$

Let X be a discrete random variable with support

$$S = \{-1, 0, 1\}$$

and

$$P(X = 0) = \frac{1}{3}.$$

Which of the following statements are true?

A. $E(X) \leq \frac{2}{3}$

B. $E(X^2) = \frac{2}{3}$

C. $E(|X|) = \frac{2}{3}$

D. $\text{Var}(X) > \frac{2}{3}$

Let A and B be two events.

Which of the following statements are **necessarily correct**?

A. $P(A \cup B) \leq P(A) + P(B)$

B. $P(A \cup B) \geq P(A) + P(B)$

C. $P(A \cap B) = P(A)P(B)$

D. $P(A \cap B) \geq P(A) + P(B) - 1$

For $n > 1$, let $X_1, X_2, \dots, X_n$ be random variables such that

- $E(X_i) = 0$ for all i
- $E(X_i^2) = 1$ for all i
- $E(X_i X_j) = \rho$ for all $i \neq j$

Which of the following statements are true?

- A. $\rho = 0$ if and only if $X_1, X_2, \dots, X_n$ are independent
- B. $\text{Var}(X_1 + X_2 + \dots + X_n) = n$ if and only if $X_1, X_2, \dots, X_n$ are independent
- C. $\text{Var}(X_1 + X_2 + \dots + X_n) = n$ if and only if $X_1, X_2, \dots, X_n$ are pairwise independent
- D. $\text{Var}(X_1 + X_2 + \dots + X_n) = n$ if and only if $\rho = 0$

Suppose X is a real-valued random variable. Which of the following values **CANNOT** be attained by $E[X]$ and $E[X^2]$, respectively?

- A. 0 and 1
- B. 2 and 3
- C. 1/2 and 1/3
- D. 2 and 5

The time between successive failures (in hours) of a side discharge loader operating in a mechanised underground coal mine are as follows:

62, 58, 54, 50, 52, 60, 58, 57, 50, 53

If the failure data follow an exponential distribution, then reliability of the equipment for a period of 50 hours is

- A. 0.25
- B. 0.40
- C. 0.60
- D. 1.00

A determinant is chosen at random from the set of all determinants of order 2 with elements 0 or 1 only. The probability of choosing a non-zero determinant is

- A. $3/16$
- B. $3/8$
- C. $1/4$
- D. none of these

Let X and Y be continuous random variables with joint probability density function

$$f(x, y) = \begin{cases} 4e^{-2y}, & \text{for } x \text{ positive and less than } y, \ y \text{ positive,} \\ 0, & \text{otherwise.} \end{cases}$$

Find the value of $E(X \mid Y = 2)$.

Let X and Y be random variables with cumulative distribution functions (CDFs) F and G , respectively.

Assume that F is continuous and, by the definition of a CDF, G is right-continuous (hence $G(-x)$ is left-continuous).

Which of the following functions of x are *necessarily* cumulative distribution functions (CDFs)?

- A. $1 - G(-x) + F(x)$
- B. $1 - F(-x) + G(x)$
- C. $F(x)G(x)$
- D. $(1 - F(x))(1 - G(x))$

Let X_1, X_2 be i.i.d. random variables which are uniformly distributed on the interval $(0, \theta)$.

Consider the problem of testing

$$H_0 : \theta = 1 \quad \text{versus} \quad H_1 : \theta = 2.$$

Which of the following tests have size ≤ 0.5 and power ≥ 0.5 ?

- A. Reject H_0 if and only if $\max\{X_1, X_2\} \geq 1.6$
- B. Reject H_0 if and only if $\min\{X_1, X_2\} \geq 1$
- C. Reject H_0 if and only if $X_1 \geq 0.4$
- D. Reject H_0 if and only if $X_1 - X_2 \geq 0$

The joint probability density function of the random variables X and Y is given by

$$f(x, y) = \begin{cases} \frac{6}{5}(x + y^2), & 0 \leq x \leq 1, 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find the covariance $\text{Cov}(X, Y)$.

- A. -0.01
- B. 0
- C. 0.01
- D. 0.02

In an experiment, a fair die is rolled repeatedly until two sixes are obtained in succession. What is the probability that the experiment ends exactly on the fifth trial?

- A. $125/6^5$
- B. $150/6^5$
- C. $175/6^5$
- D. $200/6^5$